

Question	Answer	Marks
3(a)	thermal energy per unit mass (to cause change of state)	B1
	(thermal) energy required to cause change of state between solid and liquid without change in temperature	B1
3(b)	<u>change</u> in volume is greater for vaporisation	B1
	<i>either:</i> increase in volume so work is done against the atmosphere <i>or:</i> work done against atmosphere so more thermal energy supplied	B1

Question	Answer	Marks
3(c)	$Q = mc\Delta\theta$	C1
	evidence of correct application of conservation of thermal energy [e.g. loss in energy of water as it cools = gain in energy of original ice as it warms + gain in energy of ice as it melts]	C1
	$[(34 - \theta) \times 132.4 \times 4.19] = [(\theta \times 15.6 \times 4.19) + (18 \times 15.6 \times 2.11) + (15.6 \times 336)]$	C1
	temperature = 21 °C	A1